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An Investigation into Gust Mitigation Strategies for Small Fixed-Wing Aerial Vehicles

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Abstract

Micro Air Vehicles (MAVs) and other small scaled drones have the potential to revolutionize agriculture, logistics, defense, and structural inspection tasks. The flying machines operate in the atmospheric boundary layer (ABL) which is characterized by having lower wind speeds but higher turbulence intensities compared to the flight zones of planes. The high turbulence intensities make it hard for MAVs to maintain a smooth flight and often lead to collisions. Birds also operate in the ABL and are very effective in mitigating gusts. This report takes inspiration from nature's original flyers and designed a bird-inspired wing with leading edge pressure sensing abilities. The 3D-printed experimental setup contains 11 BMP280 pressure sensor, 7 on top and 4 on the bottom, all equally distributed around 30% of the leading edge. The array detects the creation of lift by the wing and categorizes three different types of generated turbulence. Biggest differences are seen with horizontal and vertical turbulence variations.

Keywords: MAV, Gust Mitigation, Drone, ABL, Pressure Sensors

Acronym Definitions

MAV Micro-Air Vehicle

ISR Intelligence, Surveillance, and Reconnaissance

ABL Atmospheric Boundry Layer

AoA Angle of Attack

MISO Micro In Sensor Out

MOSI Micro Out Sensor In

CNS Central Nervous System

CFD Computational Fluid Dynamics

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1

Introduction

Micro-air vehicles (MAV), due to their small size, light weight, low cost, flexibility, and camouflage abilities are ideal for Intelligence, Surveillance, and Reconnaissance (ISR) missions as well as civilian applications such as logistics and fast organ transport in remote hospitals [1]. Due to the variety of possible applications, MAVs have thus gained a lot of research interest in the last 25 years [2–4].

Maintaining a smooth flight path through turbulent conditions is a significant challenge halting a wider adoption of MAVs. ISR missions and other MAV applications predominantly occur at low altitudes in the atmospheric boundary layer (ABL). Unlike higher atmospheric regions, characterized by having fast wind speeds but low turbulence intensities, where planes fly, the ABL has lower wind speeds and high turbulence intensities, visualized by the Fig 1.1. The ABL is the lowest part of the atmosphere, directly interacting with Earth's surface. The layer characteristically has large buildings which create strong eddies leading to turbulent zones. Due to the difference flying terrain, MAVs and other drones operating in the ABL should be designed differently to planes [2, 5].

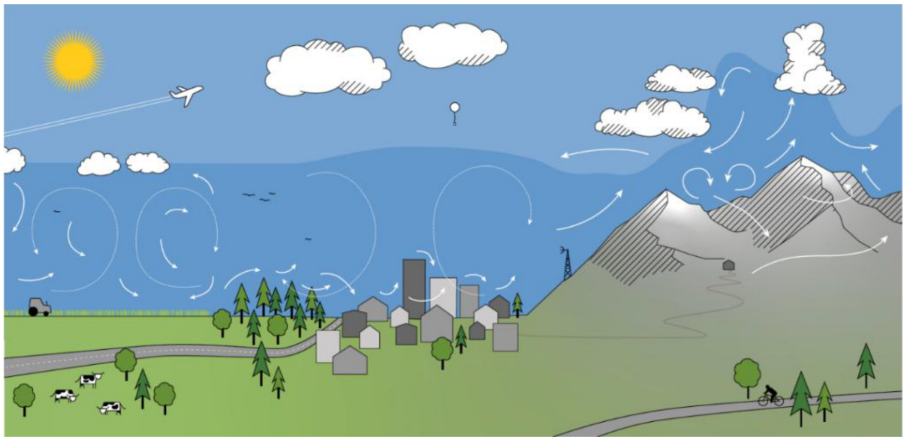


Figure 1.1: A visualization of the ABL taken from [6] and adapted from "Ideal and Real Atmospheric Boundary Layers" by Rotach and Holtslag. The left of the figure depicts an ideal scenario of the lower atmosphere interacting with a smooth surface while the more turbulent eddies towards the center and right of the figure show the more common un-ideal turbulent zones that MAV find themselves often navigating through.

Unsteady gusts degrade MAV performance through their effect on the wing's aerodynamics. Abrupt disturbances in airflow change wings effective angle of attack (AoA) to a point where flight becomes unsteady. As the AoA exceeds the stall angle, the boundary layer separates from the wings surface which reduces the lift coefficient and increases the wing's drag. The consequent loss of control manifests as unpredictable flight, degraded trajectory tracking, and increased possibility of collision. Since these disturbances occur on timescales comparable to the vehicle's own response, the rate at which a gust

can be detected and countered becomes the limiting factor on steady flight [3].

Current inertia-based attitude control systems are challenged by slow response times. Inertia sensors measure the response of the vehicle rather than the disturbance itself [5]. This detection lag leaves little time for corrective action and, without a suitable control strategy, MAVs often fail in turbulence [4, 7]. Achieving robust attitude control therefore requires sensing the disturbance before it perturbs the vehicle.

Birds share many of the same physical characteristics that make MAVs susceptible to turbulence intensities present in the ABL, yet they are often able to glide through them unfazed. [8] Thus, studies have considered the mechanisms that allow birds to glide through turbulent regions, searching for inspiration for control systems that enhance the steadiness of MAVs [5, 7, 8]. Birds utilize thousands of low resolution biological sensors throughout their wings to obtain a full aerodynamic understanding of their surroundings and respond to any incoming gusts [3]. Sensors, which can detect pressure variations over the wing can replace conventional inertia-based sensors for robust attitude control, as seen with birds. Inertia based sensors only provide the controller with information about the induced rotation of the drone. It would be preferable to instead predict perturbations induced on the drone by a gust in order to provide strategies to mitigate an induced rotation. Fly-By-Feel is a mechanism adopted from natures original fliers and uses an array of sensors placed around a vehicle's wing to assess its' flight condition.

This report focuses on using BMP280 absolute pressure sensors distributed around 30% of the leading edge of the Eppler 423 airfoil to harness incoming pressure information while considering attached flow [9]. Ultimately, the goal is to obtain aerodynamic information about the surroundings of the airfoil before the aerodynamic forces induce a moment and rotate the wing. This set of pressure data, categorized as phase-advanced information, could allow the microprocessor to use a control algorithm to counteract the predicted perturbations. Compared to birds, who have hundreds of biological pressure sensors on their wings, MAVs are limited to having a smaller number of such sensors due to their size, weight, and power contribution [5]. This experiment uses 11 BMP280 sensors, 7 on top and 4 on the bottom, wired to the Teensy 4.1 micro controller and embedded in a custom Eppler 423 wing to determine whether the manufactured setup can: (a) collect aerodynamic data about the incoming flow and distinguish between different turbulence levels, and (b) provide phase-advanced pressure measurements that indicate changes in the incoming flow or angle of attack before the airfoil's full aerodynamic response is established.

The report limits the exploration to focus on a two-dimensional airfoil and only consider the angle of attach variation due to the incoming gust. This two-dimensional treatment isolates the angle-of-attack variation as the dominant source of perturbation, allowing the pressure–AoA relationship to be characterized without the complex effects of three-dimensional flow. Spanwise effects and the broader pitch–yaw coupling are left outside the present scope and identified as directions for future work.

To conduct this investigation, the report will begin with a theoretical overview of the aerodynamic principles needed to analyze the collected data set and the electronic theory behind pressure sensors in Chapter 2. The report will follow with a review of current research surrounding bio-inspired MAV turbulence mitigation in Chapter 3. A description of the workings of the electronic components used in the experimental setup will follow in Chapter 4. An experiment will then be set up to test the hypothesis, with findings presented in Chapter 5 and discussed in the discussion Chapter in 6.

2

Theory

This project, by attempting to answer the question of how an airfoil can be designed to detect incoming turbulence, sits at the nexus point of aerodynamics and electronics. The following section provides the reader with a concise description of the theoretical concepts used in the design and data collection sections, seen in Chapters 4 and 5. The first part of this section covers the aerodynamic equations and ideas that describe the development of flow, lift, and the pressure coefficient to understand what information is key to harness from the pressure sensing array. The second part outlines the key electronic communication protocols used in the investigation, covers the working of the sensors used and concludes with a description of how an incoming gust is converted into electrical information and processed on the Teensy 4.1 microcontroller.

2.1 Fundamental Aerodynamic Concepts

In order to understand what will lead to the drone's disturbance during flight, it is first crucial to understand what causes lift and keeps the MAV hovering. The starting point for this is Bernoulli's principle, which ties together the pressure and the speed of a moving fluid. Under the assumption of an ideal flow, which is a incompressible, non-viscous, steady, and irrotational flow, traveling along a streamline, the principle tells us that the static pressure and the dynamic pressure of the flow must always add up to the same constant value. This is seen by Eq. (2.1).

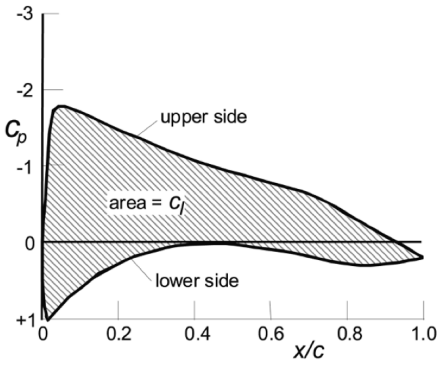
$$p + \frac{1}{2}\rho V^2 + \rho g z = \text{constant}, \tag{2.1}$$

where p is the static pressure [Pa], ρ the fluid density [$\mathrm{kg\,m^{-3}}$], V the local flow speed [$\mathrm{m\,s^{-1}}$], g the gravitational acceleration [$\mathrm{m\,s^{-2}}$], and z the height [10]. Over the small vertical extent of an airfoil the gravitational term $\rho g z$ barely changes and can be dropped, leaving the balance between static and dynamic pressure seen bellow with Eq. (2.2).

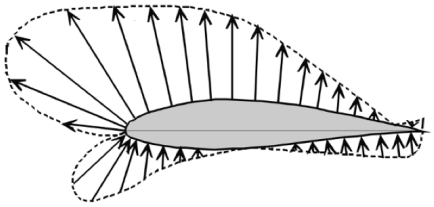
$$p + \frac{1}{2}\rho V^2 = \text{constant}. \tag{2.2}$$

Wherever the flow speeds up the pressure must drop, and wherever it slows down the pressure must rise, so that the sum stays fixed [10]. It is this inverse relationship between speed and pressure that ultimately gives rise to lift.

Lift is generated on an airfoil due to the difference in pressure between the top and bottom of the shape. The phenomenon can be explained using the Kutta-Zhukvosky Lift theorem under the assumptions of ideal flow, and Bernoulli's principle [10]. Due to the shape of an airfoil its presence in a



(a) Chordwise pressure coefficient



(b) Relative pressures normal to the aerofoil

Figure 2.1: [11] Visualization of two rendering methods of the pressure distribution around a airfoil. Image b) shows arrows which represent the relative pressure normal to the airfoil surface. The point at which the arrows separate between point away and pointing towards the airfoil is called the stagnation point and it changed depending on the AoA.

flowing liquid makes the flow wrap around it's geometry, speeding it up on the top and slowing down on the bottom. This wrapping is quantified as the circulation, Γ, [m^2s^{-1}] and is the line integral of velocity around a closed loop. Γ encapsulates the net turning of flow around an object and, for this application, can be thought of as how strong the wing is 'pulling' airflow around itself. The Kutta-Zhukvosky Lift Theorem, analytically described by Eq. (2.3), equates the lift per unit span,L, [Nm^{-1}], to the product of the fluid desnity, ρ, [kgm^{-3}], the free stream velocity, U, [ms^{-1}], and Γ [9]. This circulation produces the pressure distribution over the airfoil that gives rise to lift and determines its lift coefficient.

$$L = \rho U \Gamma, \tag{2.3}$$

The lift coefficient, C_L, is a dimensionless parameter which quantifies the effectiveness of a body in producing lift relative to the surrounding air pressure and surface area. It is given by Eq (2.4) below next to the pressure coefficient, C_p, formula. L referring to the lift force [N], S to the reference surface area [m^2], p the pressure [Pa] and, p_∞ the fully developed pressure [Pa].

$$C_L = \frac{L}{\frac{1}{2}\rho V^2 S}, \qquad C_p = \frac{p - p_\infty}{\frac{1}{2}\rho V^2} \tag{2.4}$$

Measurements of pressure around an airfoil can be converted into a distribution of the pressure coefficient as visualized in figure 2.1. The lift coefficient can then be determined by the derivation below taken from [11].

The aerodynamic force normal to the chord arises from the pressure acting on the airfoil's surface.

Integrating the pressure over the lower and upper surfaces gives the normal force per unit span n. At small angles of attack the lift per unit span l is well approximated by this normal force:

$$l \approx n = \int_{\ell} p_{\ell}\,\mathrm{d}x - \int_{u} p_u\,\mathrm{d}x, \tag{2.5}$$

where the subscripts ℓ and u denote the lower and upper surfaces respectively. The two integrals have opposite signs because pressure on the lower surface pushes the airfoil upward, while pressure on the upper surface pushes it downward.

Equation (2.5) contains absolute pressures, whereas only the pressure difference relative to the surroundings can produce a net force. The constant ambient pressure p_∞, integrated over both surfaces of a closed contour, produces no net force, since both integrals run over the same chordwise limits:

$$\int_{\ell} p_\infty\,\mathrm{d}x - \int_{u} p_\infty\,\mathrm{d}x = 0. \tag{2.6}$$

Subtracting Equation (2.6) from Equation (2.5) leaves the lift unchanged while expressing it in terms of pressures:

$$l \approx n = \int_{\ell} (p_{\ell} - p_\infty)\,\mathrm{d}x - \int_{u} (p_u - p_\infty)\,\mathrm{d}x. \tag{2.7}$$

Finally, dividing both sides by the dynamic pressure $q_\infty = \frac{1}{2}\rho V^2$ and the chord length c non-dimensionalises the equation. The left-hand side becomes the lift coefficient $c_l = l/(q_\infty c)$, each pressure difference on the right-hand side becomes a pressure coefficient $C_p = (p - p_\infty)/q_\infty$, and the integration variable becomes the normalised chord position x/c:

$$c_l = \int_0^1 C_{p,\ell}\,\mathrm{d}(x/c) - \int_0^1 C_{p,u}\,\mathrm{d}(x/c) = \int_0^1 (C_{p,\ell} - C_{p,u})\,\mathrm{d}(x/c). \tag{2.8}$$

The lift coefficient is therefore given directly by the area enclosed between the lower- and upper-surface pressure coefficient curves in figure 2.1.[11]

2.1.1 Roll rates

The rotations of flying vehicles are described by their yaw, pitch, and roll. These are rotation directions and are visualized with fig 2.2.

Roll dynamics are particularly important for small fixed-wing drones because atmospheric turbulence generates large differences in lift between the left and right wings. This large lift difference can produce rolling moment, physically rotating the drone. Pitch disturbances are typically moderated by the aircraft's longitudinal static stability and horizontal tail. Yaw disturbances primarily affect heading (the direction of the nose of the aircraft) and are generally weaker because turbulence produces smaller side-force asymmetries than lift asymmetries [13]. Consequently, roll is often the dominant attitude response to gusts and becomes the limiting axis for flight stability in MAVs.

Mohamed et al. showed showed that the roll rate time constant of a wing scales with the square root of the it's wingspan [4]. This means that for a 1/10-scale vehicle both the frequency of the roll response and the amplitude of the roll rate increase by a factor of $\sqrt{10}$ making the smaller wing react to a sharp gust both faster and more violently. Below is a shortened derivation of his findings crucial

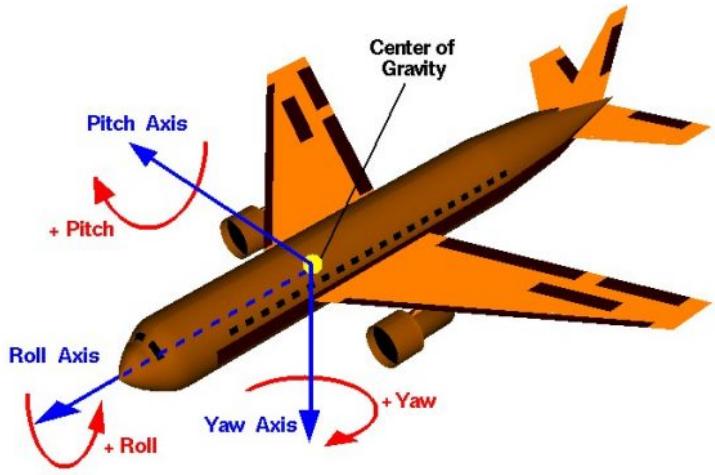


Figure 2.2: The yaw, pitch, and roll rotation directions of a flying vehicles [12].

to the understanding of why MAV have such trouble with turbulent flight.

We start with the roll damping stability derivative L_p. L_p is the result of the opposing moment to a induced roll moment from a incoming gust [14]. It is analogous to shock absorbers of a car, resiting the incoming force. A larger L_p will dampen a induced roll more quickly. It is given by the equation below:

$$L_p = \frac{qSb^2C_{l_p}}{2I_{xx}U} \tag{2.9}$$

where q is the dynamic pressure, S is the wing area, b^2 is the wingspan, C_{l_p} is the (non-dimensional) roll damping coefficient, I_{xx} is the moment of inertia about the roll axis, and U is flight velocity. Physically, a larger L_p means the aircraft damps off a roll disturbance more quickly.

Under simplifications [5, 14], the roll mode time constantan be approximated as the inverse of L_p as:

$$\tau_r = \frac{-1}{L_p} \tag{2.10}$$

$$\tau_r \approx \sqrt{b} \tag{2.11}$$

Eq. (2.10) assumes that wingspan, b, is the governing length scale, and that everything else scales geometrically from it. The author concludes the square root of b scaling by using proportionality relationships of the other variables in Eq. (2.9) with respect to b. The full derivation and reasoning for the proportionalities can be found in Section 7.1.

2.1.2 The Küssner problem

Unlike steady aerodynamic theory, which assumes that lift changes instantaneously with the surrounding flow, the Küssner solution captures the finite time required for aerodynamic loads to develop as a gust passes over the wing. This transient response is particularly important for micro air vehicles (MAVs), where rapid gust encounters occur on timescales comparable to the aircraft's dynamic response.

The Küssner problem considers transient lift developing on a thin airfoil as it penetrates a sharp-edged vertical gust [13]. Here, a vertical gust refers to an upward flow disturbance encountered by

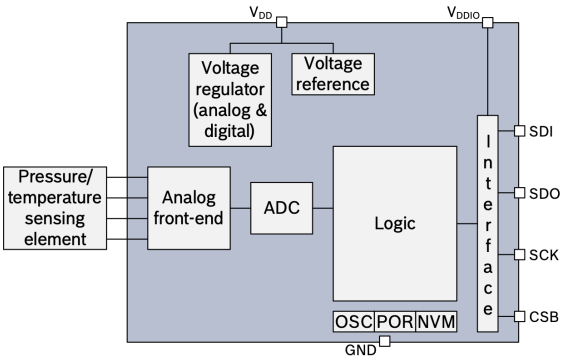


Figure 2.3: BMP280 block diagram depicting the signal pipeline from detection to digital output, taken from the BMP280 datasheet [17].

the airfoil. It was first posed by Küssner (1935) and solved analytically by von Kármán and Sears (1938) [13]. Küssner's airfoil meets the gust gradually, so the quasi-steady incident force builds up progressively as the airfoil immerses itself in the upwash. From my understanding I find this analogous to static and dynamic friction, where only after a set force will the object start moving. After a build of forces the friction gives in. Similarly here the aerodynamic forces need to build upon the wing before they start moving. Mathematically it becomes fully immersed at a travel of $s = 2$ semi-chords. A semi-chord being defined as half of the chord. The semi-chord is a unit in turbulent aerodynamics it provides a non dimensional distance, with s = Ut/b. b referring to the full chord length, U the wind speed, and t the time elapsed. The upwash velocity w_g seen by the airfoil is defined relative to the leading edge as a step function given by Eq. (4) in the appendix.

The Küssner approach, while expensive in computation time, is a good theoretical tool to explain the small time scale that lives between the gust facing the wing and the gust making the wing move. It is mathematically described in the appendix in section 7.1

2.2 Communication Protocols and Embedded Implementation

Transferring information between a sensor and a processor requires a communication protocol. The protocols relevant here are serial, meaning data is sent one bit at a time over a wire [15]. A device with separate transmit and receive wires can do both simultaneously (full-duplex), while a device sharing a single data wire must alternate between transmitting and receiving (half-duplex). The peripheral used throughout this project, the BMP280 barometric pressure sensor, supports both modes.

The BMP280 is an absolute pressure sensor as it measures pressure relative to a perfect vacuum using a silicon diaphragm with embedded resistors. Mechanical deformation of the diaphragm changes the resistance, producing a differential voltage in the microvolt range [16]. This signal is amplified, improving the signal-to-noise ratio, and converted to a 20-bit raw digital value which must then be transferred to the processor [17]. The pipeline is shown in figure 2.3.

The BMP280 offers two interfaces: half-duplex I^2C and full-duplex SPI [17]. I^2C uses two wires, a serial clock line (SCL) and a serial data line (SDA). Each transaction begins with the controller sending a 7-bit peripheral address indicating a read or write, the addressed peripheral acknowledges, data is transferred, and the controller ends the transaction with a stop bit. SPI uses four wires: an SCL

for synchronisation, a MISO (Micro In Sensor Out)[1] line for data into the controller, a MOSI (Micro Out Sensor In) line for data out of the controller, and an active-low chip select (CS). Because each peripheral has its own CS line, multiple sensors can share one SPI bus, which made SPI favourable for wiring several sensors to the Teensy 4.1.

The protocols differ substantially in achievable data rates. For the BMP280, I^2C supports clock speeds up to 3.4 MHz, whereas SPI accepts up to 10 MHz [17]. Moreover, every I^2C transaction carries addressing and per-byte acknowledgment overhead on a half-duplex line, while SPI transfers data with no addressing, no acknowledgment bits, and full-duplex operation. SPI is therefore the faster protocol, at the cost of one additional wire per peripheral. The following section explores how these communication speeds become the determining factor for phase-advanced gust detection.

[1] MISO originally stood for Master In Slave Out, and likewise MOSI for Master Out Slave In. In 2020 the Open Source Hardware Association (OSHWA) proposed replacing these with COPI/CIPO, using controller and peripheral, but two years later realised the latter is an offensive term in Polish and changed the terminology again to PICO/POCI [18, 19]. This report uses MISO/MOSI with non-profane acronym definitions, as many older sensors and papers use the original terminology.

3

Literature Review

Over the past 25 years, MAVs have attracted considerable research interest owing to their wide range of potential applications. [2–4, 8, 20] This literature study opens with an overview of Mohamed et al. research at the RMIT university on MAV and gust control. This research pool from RMIT is used in the report as a theoretical bedrock and is what a lot fo the concepts and ideas are based off. The chapter then follows with a overview of Cheney et al. publication on bird wings being used as a passive suspension system in response to upwards gusts. This study is used in the report as a backdrop for one of the potential use cases of this research. Being able to detect oncoming upwards or downwards gusts in time for mechanical wings to react.

3.1 Background Research

The School of Aerospace, Mechanical and Manufacturing Engineering at the RMIT University in Melbourne has contributed significantly to the field of MAV flight control. Several publications by Mohamed et al., established much of the conceptual framework that this report builds upon. The groups central thesis is that the difficulty of flying a MAV steadily through turbulence is not a single problem but an array of cascading problems. The vehicle's miniature scale coupled with the other problems that arise from making things small; energy storage, sensing mechanisms, and other actuator problems make it hard to design efficient MAVs. The group uses on biomimicry, which is the technique of studying animals and copying their features for an engineering purpose, to come up with novel solutions.

In the review paper [8] the research group concluded that turbulence in the atmospheric boundary layer can severely degrade attitude stability and mission effectiveness and identified both passive and active approaches for disturbance rejection. Building on this, the group demonstrated that conventional inertia-based sensors limited MAV flight performance in turbulence due to their inherently reactive nature.[4]. To address these limitations the authors proposed the gust perturbation process and discussed the advantages of using 'phase advanced sensors'. Turning to natures original flyers, the group investigated a range of biological mechanoreceptros and concluded that pressure and flow sensors were most promising candidates for the 'phase advanced sensors' the group had in mind. **??**

More recently, Mohamed and co-workers produced a review paper [8] with the intent of understanding why birds exhibit remarkable steadiness under conditions that severely challenge small uncrewed aerial vehicles. The article found that birds use a large array of different sensors all tuned to perform a specific task and concluded that to achieve a 'fly-by-feel' control protocol research should focus on incorporating several aerodynamic sensors within a MAV.

A novel bird gust detection mechanism has been proposed by Cheney et al. [20] (2020). In collaboration between the University of Bristol, the Royal Veterinary College, and Dongguan University of Technology, investigated how bird wings passively reject gust disturbances. Using a captive-bred female barn owl (*Tyto alba*) flying through a controlled indoor corridor, the researchers combined high-speed motion capture, three-dimensional surface reconstruction, CT scans, and CFD simulations to analyse the owl's aerodynamic and inertial response to upward gusts.

The study found that the owl's wings acted as a passive suspension system, rotating upward about the shoulder within approximately 80 ms, before neural reflexes could respond. This motion absorbed a significant portion of the gust energy, reducing the impulse transmitted to the torso by approximately 32%. Subsequent aerodynamic changes in wing pitch and shape further reduced the peak vertical impulse by approximately 42–63% compared with a rigid-wing model. The authors concluded that incorporating passive hinged-wing or suspension-like mechanisms into flapping-wing MAVs could improve gust rejection and flight stability while reducing control complexity.[20]

3.2 The Gust Perturbation Process

The gust perturbation process describes the measurable chain of events that unfolds as a turbulent gust advances over a wing. The process begins with the disturbance in the flow and ends in a deviation of the vehicle from its intended path due to an AoA change. The following framework is taken from Mohamed et al. [5], which was adapted from [4] and reproduced in figure 3.1.A perturbation in roll was chose since Mohamed et al. [5] identified it as most problematic for MAVs. The central insight of this framework is that the flow disturbance is fully present in the flow at stage one, several steps before it manifests as the vehicle motion that an inertial sensor can register. Breaking up this process into several stages before the turbulence manifests itself into the rotation of a wing or derailment of a drones trajectory allows for new ways of thinking about phase advanced control and interpreting the incoming gust before it creates a moment.

Conventional inertia-based sensors, typically present on MAVs measure the inertial response of the airframe to a disturbance rather than the disturbance itself making them inherently reactive.[5] By the time this type of sensor produces a signal, much of the perturbation chain has already run its course, leaving little time for any corrective action. Sensors that instead measure the early stages, the pressure and velocity variation across the wing, are termed *phase-advanced* sensors because they detect the disturbance before it acts on the vehicle.

Between each event there is a theoretical phase-lag, or small amount of time, and hence sensors can be put in between each section to harness information, allowing for a phase-advanced response. A sharp gust incident on the leading edge of a wing, or the airfoil in the case of this project, does not lead to a an instantaneous increase in the wings lift, but a gradual increase in lift as the gust penetrates the chord and the wake adjusts. This build up is described by the Küssner function, $\psi(s)$ and given by Equation (6). [13]

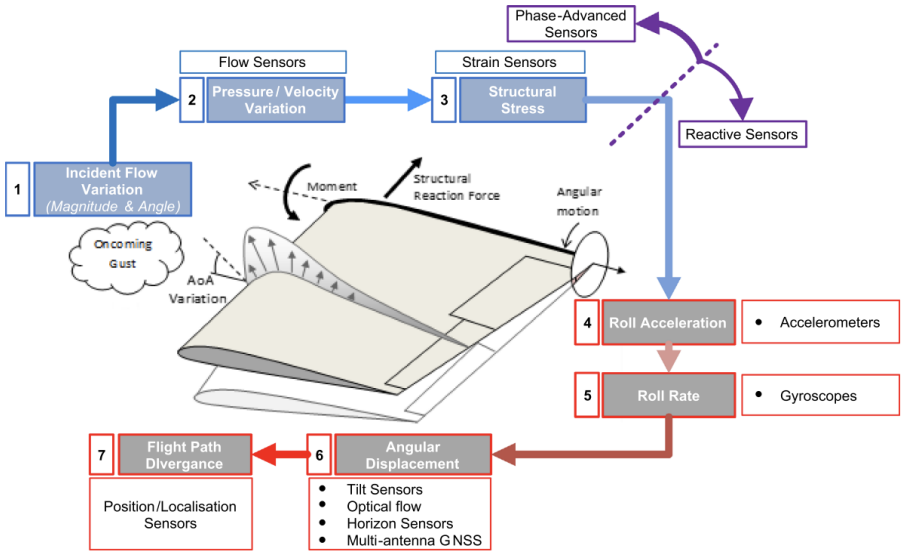


Figure 3.1: The measurable sequency of events caused by incoming turbulence over a wing and their attributed sensors taken from [5] and adapted from [4]. The process is represented to be broken up into 7 stages.

3.3 Bird Mechanoreceptors

Bird wings are filled with various mechanoreceptors which are specialized sensory receptors that convert mechanical agitation into electrical signals interpretable by the central nervous system (CNS) [21]. Situated near the feather follicles, these receptors allow birds to sense airspeed and detect both stall and turbulence. It is important to highlight the importance of the large number of sensors present, as it is hypothesised that birds use a large array of low-resolution sensors, in contrast to the engineering approach of relying on a single, highly sensitive measuring device.[8]Each mechanoreceptor type has a unique response characteristic optimised for a specific task.[22]

Out of the variety of receptors that line the bird, ranging from free nerve endings which are used to detect pain and temperature to encapsulated endings which are a class of mechanoreceptors, Herbst corpuscles are a class of mechanoreceptors and are ones known to contribute most to flight control in birds. [23] They mimic flow pressure sensors and are found in feather follicles which sense vibration.[5] Their distribution of varies between species; for example, pigeons have vibration-sensitive receptors in their primary wing feathers, while chickens mainly possess them in secondary feathers. Overall, the receptors are densely concentrated in aerodynamically important regions such as the wing's leading edge and the alula.

Hörster [24] found that Herbst corpuscles fire only after a threshold frequency and amplitude are reached. The sharp tunning to a narrow frequency band gives them high sensitivity for detecting specific airflow disturbances. Across a tested range of 50–2000 Hz, peak sensitivity occurred around 800/900 Hz, with an optimal deflection amplitude of 0.09–0.5 mm. The high firing rate at these frequencies is thought to act as a warning signal for dangerous flight conditions, potentially grounding the bird, and may also help optimize the wing's angle of attack and morphing for efficient flight.[5]The key engineering insight drawn from this biological system is not the sophistication of any individual receptor, but rather the large distributed of the array.

3.4 Research Gap

After reviewing, to the best of my abilities, the current literacy basin surrounding phase advanced gust detection for MAV flight control I realized that there was little focus on using basic, readily available pressure sensors to detect incoming gusts. Due to their relative simplicity, in terms of integration and operating physics, absolute barometric pressure sensors like the BMP280 would offer a simple solution to a complex solution. Often in science such 'simple' solutions are best suited for such 'complex' problems.

Thus, this project looks at if and how a conservative array of BMP280 pressure sensors, embedded over the top and bottom of a airfoil, can detect phase advanced aerodynamic information. If positive, the findings would aid the future MAV research and design by giving engineers a already well established technology to obtain crucial information. Due to the low costs of such sensors the research and design phase could afford to have many more iterations, often leading to more promising designs.

4

Methods

This chapter describes how the experimental apparatus was setup up to answer the reports research question. First the research question and the hypothesis are presented, secondly the way in which the research question was answered is discussed. The chapter concludes with comments on experimental techniques and data aquisition.

4.1 Research Question

To what extent can an array of basic barometric pressure sensors, distributed around the first 30% of the leading edge of a bio-inspired airfoil, resolve variations in turbulence intensity in its surroundings?

4.1.1 Hypothesis

It is hypothesized that the pressure sensor array can detect variations in turbulence intensity, to a limited extent. The sensors are placed on the first 30% of the leading edge, which is where the pressure changes the most when the incoming flow changes. Vertical turbulence is expected to be the easiest to detect, since vertical gusts change the effective angle of attack of the wing and therefore have a direct effect on the pressure near the leading edge. Since the BMP280 is a basic low-cost sensor with a limited sampling rate, the array is expected to separate rough categories of turbulence rather than small variations.

4.2 Airfoil Design

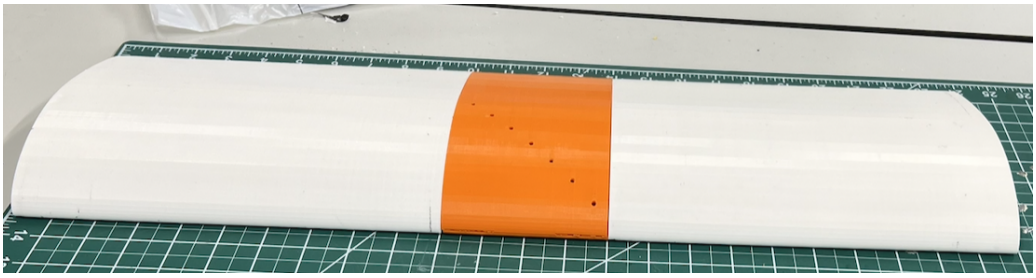
The airfoil was modeled after the Eppler 423 standard airfoil and mirrored as a *.dat* file from the UIUC Airfoil Coordinate Database. The Eppler 423 airfoil follows the requirements of bird-inspired airfoil designs, such as having a pronounced camber, high lift coefficients and a good performance at low Reynolds numbers and was thus chosen for this bird-inspired turbulence detection investigation.

The airfoil was designed using SolidWorks and 3D printed. The airfoil was 3D printed (Ultimaker S3, 20% infill density) with orange and white filament (Ultimaker PLA). The standard 'engineering design' infill density offered a sturdy structure while maintaining conservative filament usage. The design process went through several iterations and concluded on the design seen in figure 4.1a. The various prototypes can be found in the Appendix 7.1 at the end of the report.

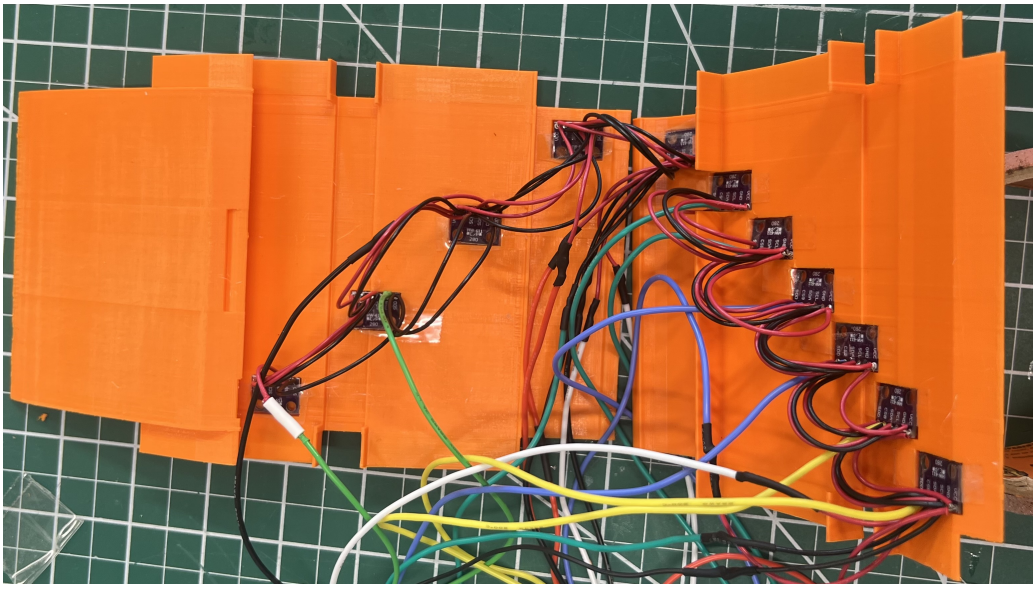
The wing structure was printed in five sections. Two outer sides of the airfoil were printed with a $200mm$ chord length and a $250mm$ span. The middle orange sections was printed in two parts, the upper and lower part, with same chord length but a $30mm$ span. The inner orange part is characterized

with having pressure taps for the sensors rather than wider openings and has the taps arranged in a diagonal cascade rather than stacked one after another. The two design choices stemmed from wanting to make the airfoil as aerodynamically smooth as possible and avoid creating wake regions behind taps or open pressure sensors.

11 Pressure sensors were used on distributed equally around the first 30% of the leading edge of the airfoil. 4 sensors were placed on the bottom and 7 on top. The leading edge was chosen both due to it experiencing the oncoming gusts first and as there was enough physical material to work with on that area. Further down the chord the airfoil got too thin to have sensor holders.



(a) The 4 parts of the setup put together.



(b) The two middle components opened apart showing the wiring of the cables.

Figure 4.1: Image a) shows the full setup spanning 43cm in length. Image b) depicts the butterfly wiring of the BMP280 sensors.

4.3 Communication Protocols

We used the BMP 280 barometric pressure sensor, the new and improved version of the original BMP 180. The sensor was chosen due to its SPI communication protocol. The BMP280 offered both I^2C and SPI communication pathways which proved crucial in the completion of the project. The sensors compact package $2.0mmx2.5mmx0.95mm$ makes it very suitable for a embedded design. The sensor comes with a relatively high accuracy when compares to other barometric pressure sensors at it's price

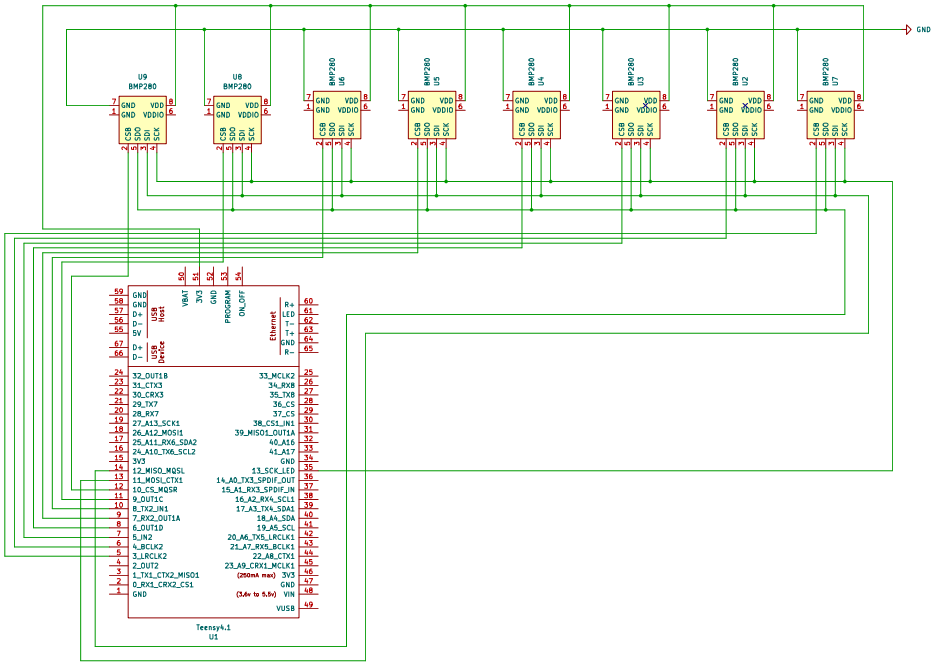


Figure 4.2: KiCad electrical drawing for the wiring of the BMP280 sensors with the Teenys 4.1 microcontroller. The diagram depicts 8 out of the 11 sensors used but the wiring for those is the same.

range, $0.12hPa$. It's main drawback is it's long measurement time, ≈ 5.5ms , and intended design for altitude control and weather monitoring rather than drone flight control[17].

The investigation began by using the BMP180 pressure sensor which only offered I^2C communication. The initial implementation of I^2C failed as the communication is limited by the number of individual addresses available on the microcontroller. The Teensy 4.1, the microcontroller used throughout the investigation, has 3 bus addresses so a more complex shared bus system using pull-up resistors and a multiplexer board had to be implemented. I could not get it to work. To read more about this communication protocol please refer to chapter 7 of [15].

The BMP 280 sensors were wired together using wire and soldered in a butterfly configuration seen in 4.1b. The sensors were wired as seen in the electronic schematic in figure 4.2.

4.4 Data Acquisition

The experimental setup used the active turbulence generation grid to generate oncoming gusts. The investgation conducted x experiments. First the ability of the wing to generate lift was calculated. This was done by starting the pressure measurements without oncoming wind and then starting the fans and having them spin to create wind traveling at 6.5ms^{-1}.

4.4.1 Active Turbulence Generation Grid (ATG)

The ATG grid consists of horizontally and vertically mounted triangular flaps. The flaps are rotated by vertically and horizontally placed motors and can rotate in unison. Gusts are created by rotating the flaps in various ways. Fig. 4.3 depicts the pulsing turbulence generation and can be done at various RPM.

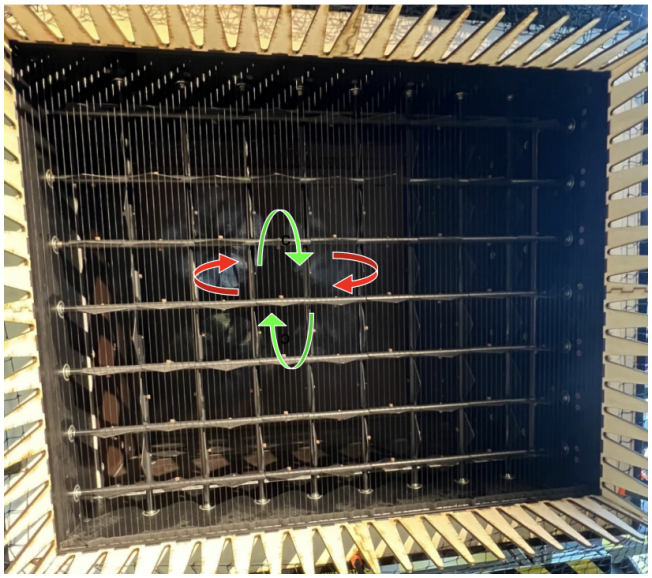
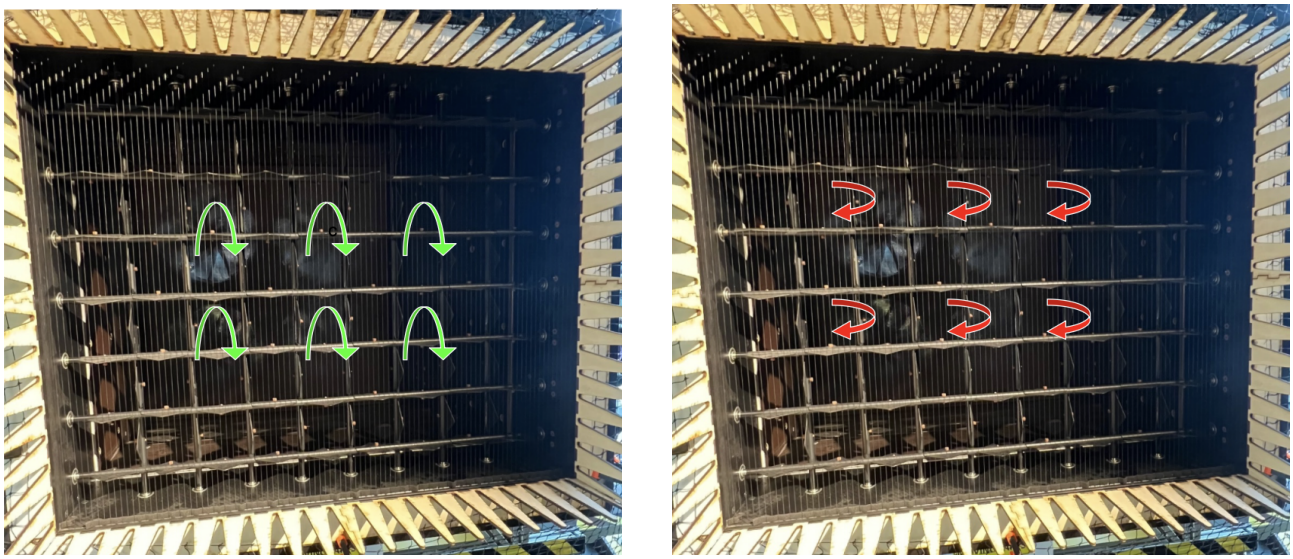


Figure 4.3: Pulsing gust generation method by the ATG.

Rotating the vertical and horizontal flaps only, while maintaining the other ones fully open is also possible and is called shear turbulence. The two types of shear turbulence are depicted in Fig. 4.4



(a) Horizontal turbulence **(b)** Vertical turbulence

Figure 4.4: Shear turbulence generation, with flaps opening 110ž measured from the min to max.

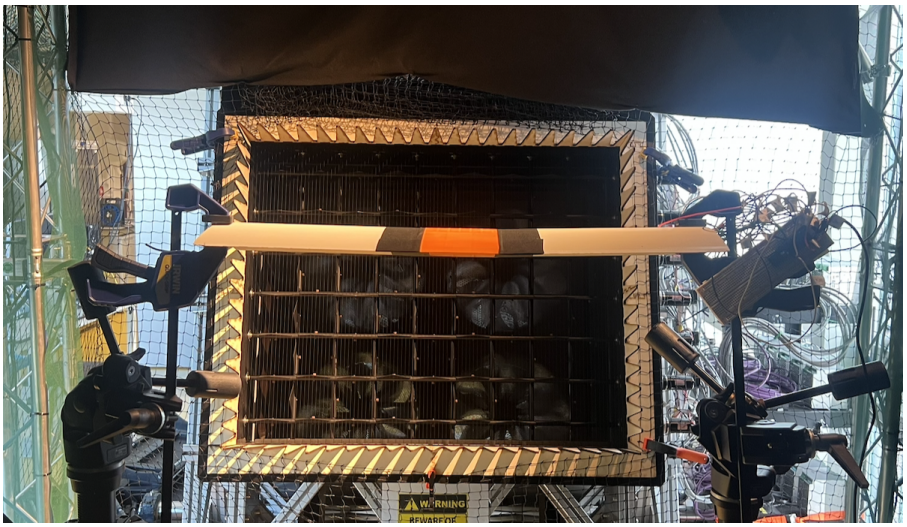


Figure 4.5: Experimental setup used throughout the investigation. The airfoil is connected to the Teensy 4.1 and data was collected using a USB extension cable so the computer did not interfere with the turbulence.

4.4.2 Experimental Setup

The experiment was carried out using a wind tunnel which ran at 6.5 ms^{-1} as that was the wind speed the birds in the research group flew in in other experiments. The wing is seen mounted using two tripods and leveling app on the iPhone 13.

4.4.3 Detection of Low and High Turbulence Conditions

The first experiment tested whether the pressure sensor array could distinguish between different incoming turbulence conditions for a static airfoil.

The airfoil was mounted at a fixed AoA 6 degrees relative to the oncoming airflow in the wind tunnel to maximize the lift coefficient. Measurements were taken using turbulence gust generation and ambient wind speed.

Turbulence was generated using a grid of moving triangular flaps that provided random, shear and pulsing turbulence, all at varying rotations per minute (RPM). Random gusts are defined as the gusts created by flaps moving in random directions at random times, generated by a pre-defined random generation algorithm, but at set RPM. Shear gusts were divided into two parts; gusts created with only the motion of the horizontally mounted flaps and gusts generated with only the motion of the vertically mounted flaps, while maintaining the non-oscillating flaps opened (parallel to the airflow). The shear gusts also varied the amplitude of oscillation, rotating a preset number of degrees from their idle horizontal or vertical position. Pulsing gusts are gusts generated by both the horizontal and vertical components of the grid rotating in unison at a preset amount of RPM. All experiments were done using the same 6.5 ms^{-1} wind speed as that is the wind speed that the Kestrels fly in during other experiments within the group.

5

Results

The results consist of absolute pressure readings from the 11-pressure sensor array used in the setup. There were 4 sensors placed on the bottom of the foil and 7 placed on the top. To quickly recognize which graph represents which sensor location a blue-red color scheme has been implemented. Figure 5.1 shows how the red and blue colour gradient has been assigned to each sensor. The gradient of blue represents bottom sensors while red indicates sensors placed on the top. The darker the colour the closer it is to the leading edge of the foil. As the report is concerned about the aerodynamic response of the airfoil rather than the time evolution the time scale of measurements was relatively unimportant and thus the results may often show varying time scales for taken measurements.

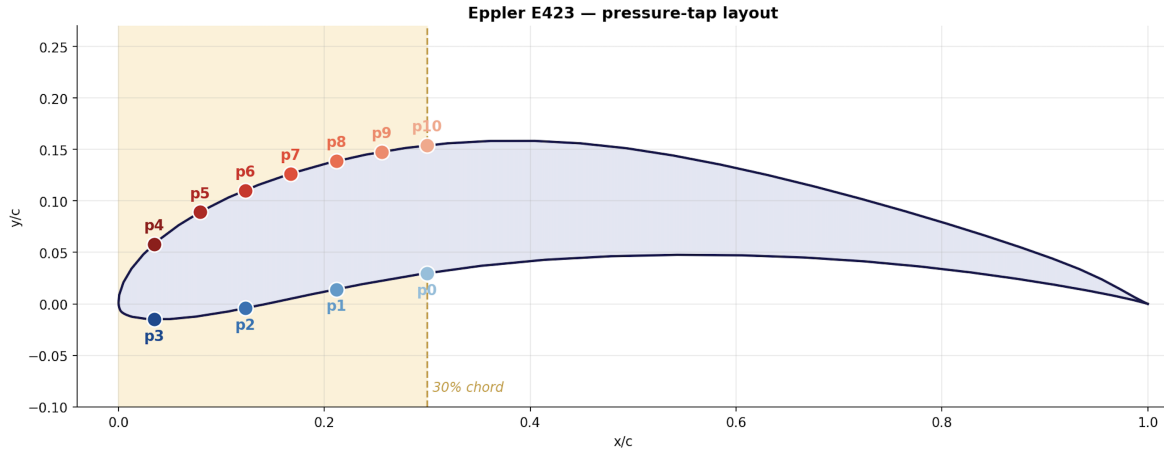


Figure 5.1: Pressure sensor distribution for the Eppler 423 airfoil used in this project. The color gradient is used to help the reader quickly visualize which graph represents which pressure tap.

5.1 Sensor Validation

The results immediately indicated that the pressure sensor array worked. The sensors not only responded to incoming wind but when graphed provided strong evidence for the formation of lift. The pressure difference due to the sudden oncoming wind is seen in. Figure 5.2 depicts the starting up and turning off the wind tunnel and this behavior is seen in the figure. The measurement in 5.2 was taken as the wind tunnel was starting and so a initial startup phase can be seen before lift is fully established. This result highlights that the setup both can both record barometric pressure data and produce lift.

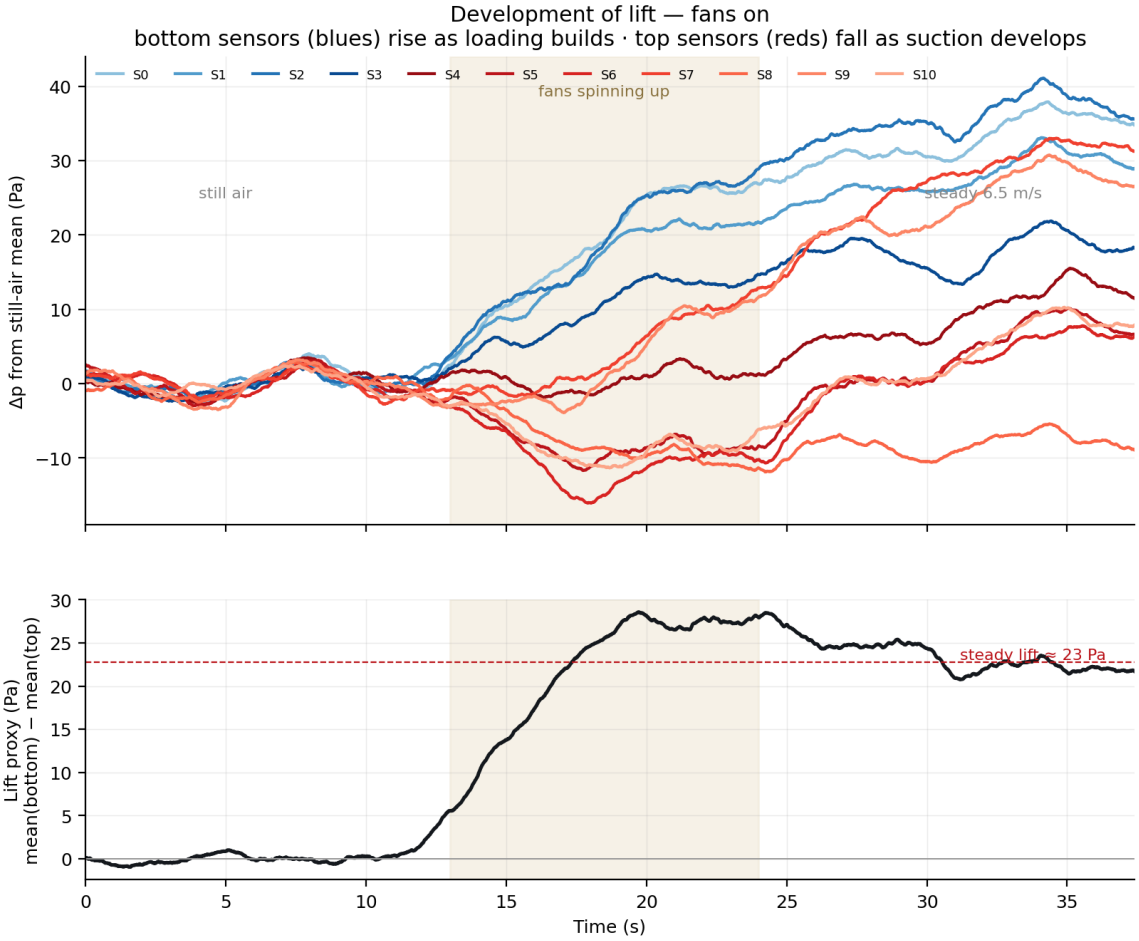


Figure 5.2: The pressure distribution of the senors varied with time. The figure notably shows the creation of lift as wind is introduced as a build up rather than a theoretical step input. The wind speed settles at approximately $6.5ms^{-1}$. Wind is then turned off again as a varied input rather than a step. The measurements are presented with respect of a baseline. The baseline is the average pressure for each sensor taken before wind was introduced and masked as 0 to homogenize each start time.

5.2 Steady Airflow

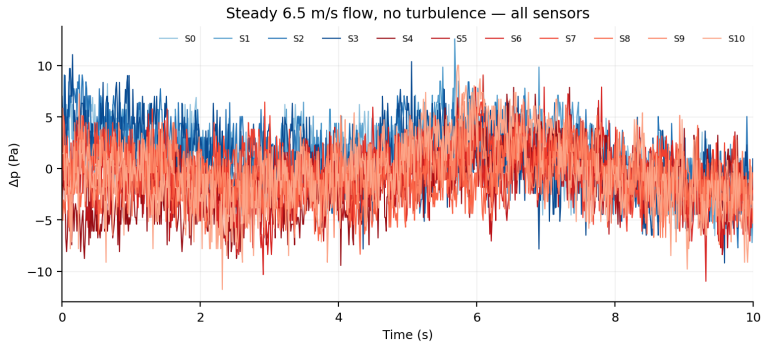
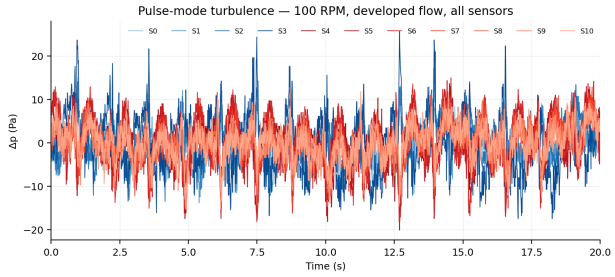


Figure 5.3: Normalized pressure data taken from BMP280 readings at 6.5m/s wind speeds over 10 seconds for a clear reading.

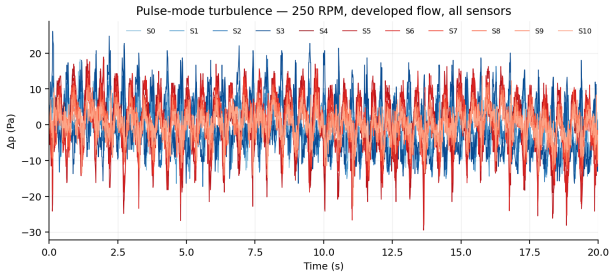
Figure 5.3 shows the baseline readings of BMP280 sensors when subject to a constant steady wind. The presure data for each sensor was normalized by taking a mean value for 10 seconds and setting that to a basline and recording deviations from it for each sensor.

5.3 Turbulence Generation

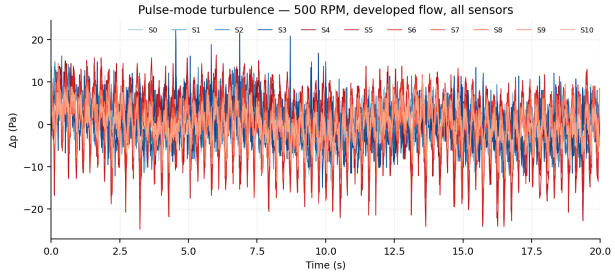
Below are the results for both pulsing and shear turbulence generation. The shear results are further split into two; horizontal and vertical shear. The 10-second time series of the pulsing turbulence generation results are seen below.



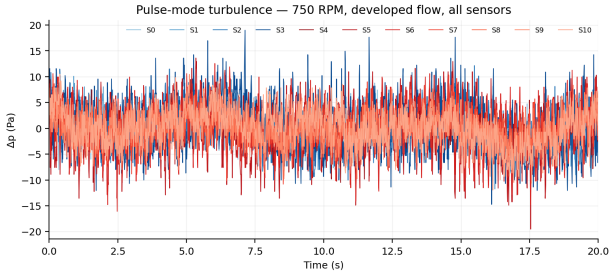
(a) 100 RPM



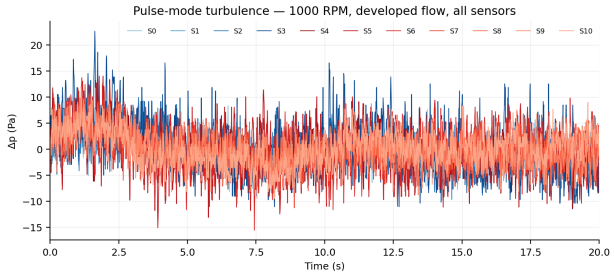
(b) 250 RPM



(c) 500 RPM

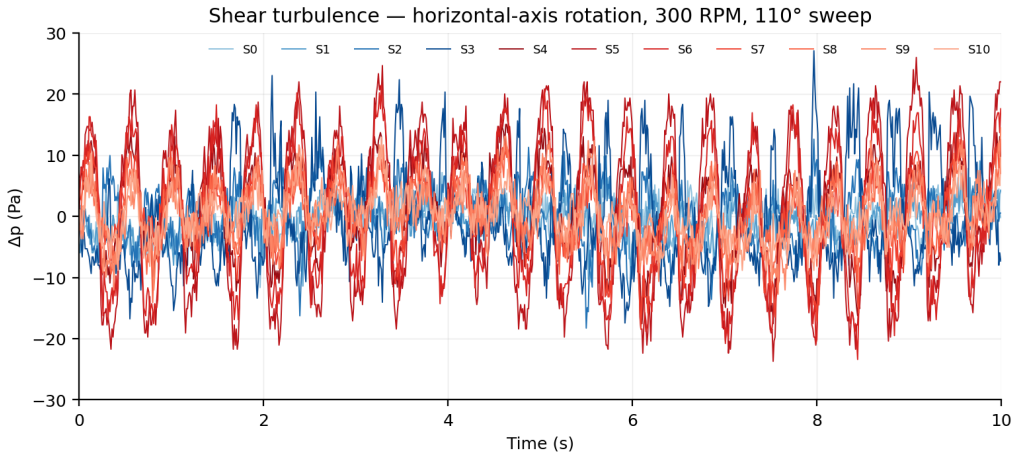


(d) 750 RPM

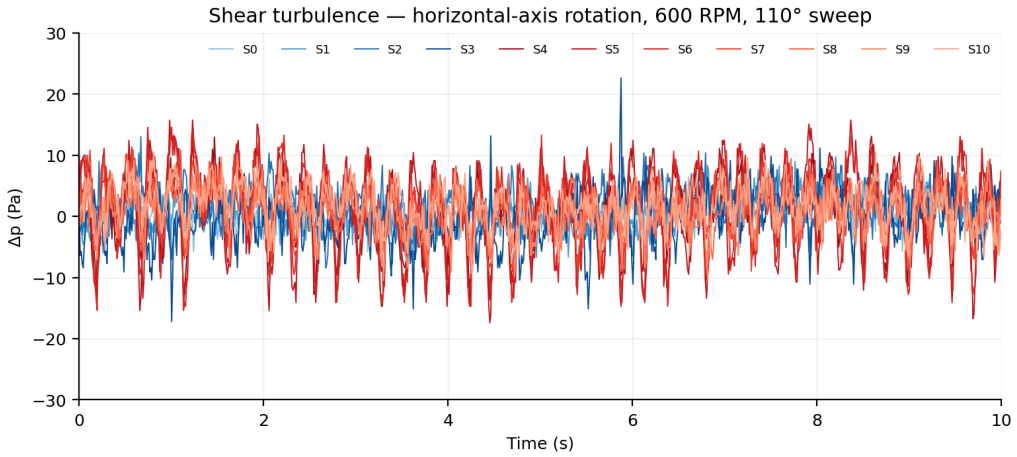


(e) 1000 RPM

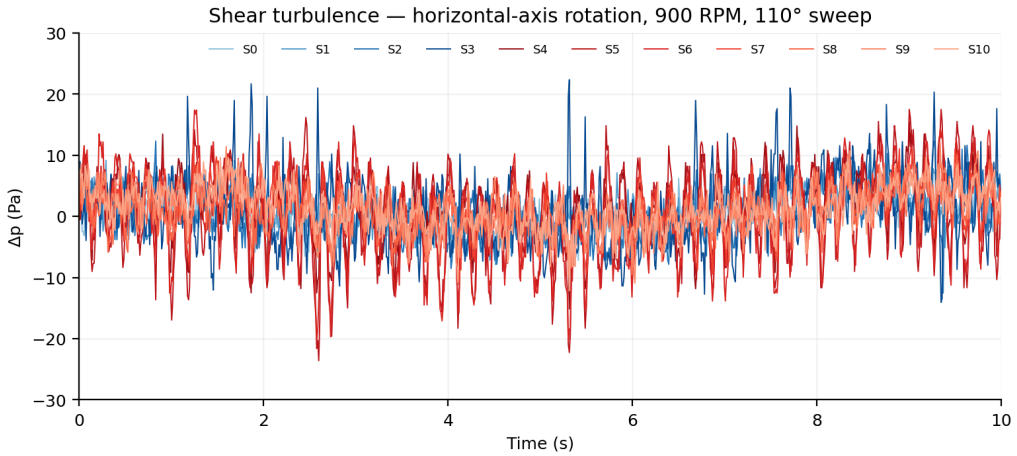
Figure 5.4: Normalized pressure at all eleven sensors under pulsing turbulence, over the first 10 s of the developed-flow window, at grid speeds of (a) 100 RPM, (b) 250 RPM, (c) 500 RPM, (d) 750 RPM and (e) 1000 RPM. The fluctuation amplitude grows with grid speed.



(a) 300 RPM

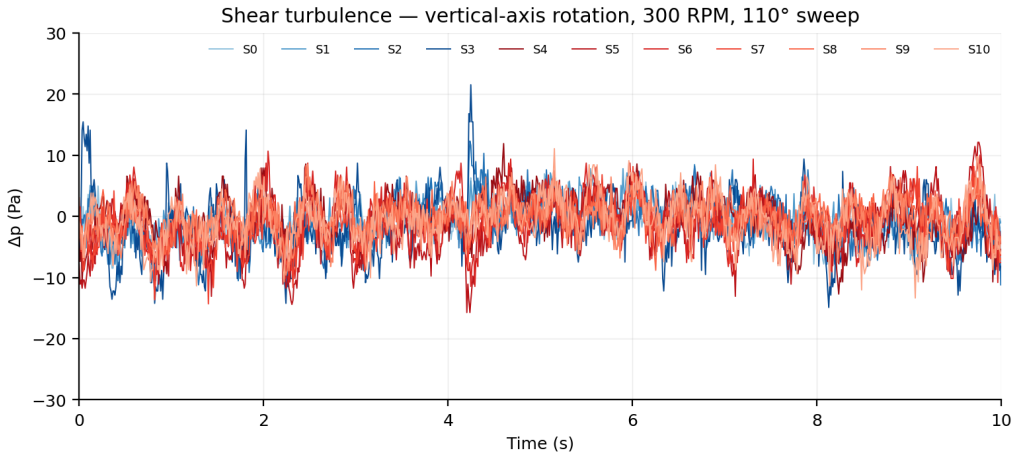


(b) 600 RPM

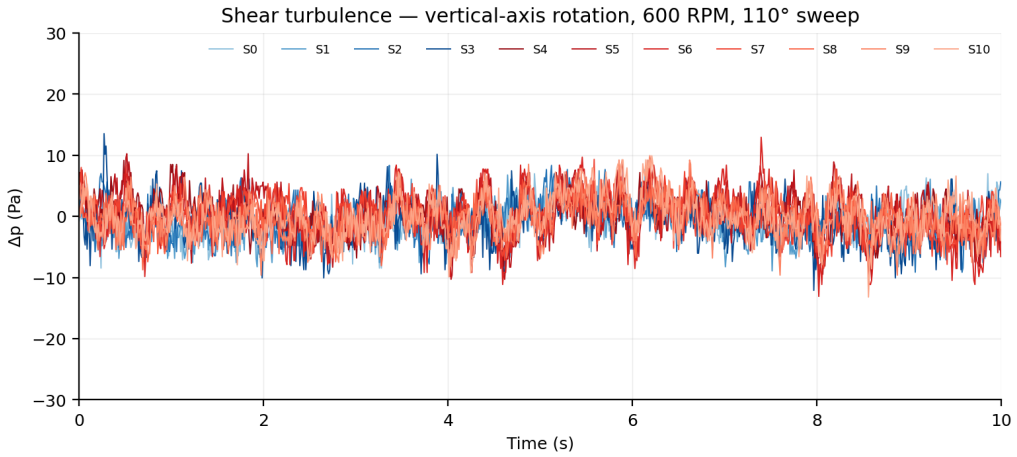


(c) 900 RPM

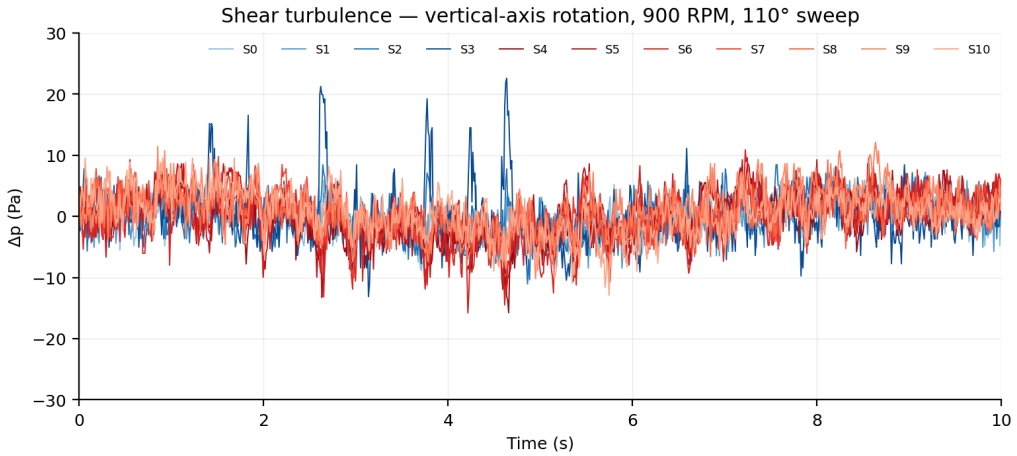
Figure 5.5: Normalized pressure at all eleven sensors under horizontal-axis shear turbulence, over the first 10 s of the developed-flow window, at grid speeds of (a) 300 RPM, (b) 600 RPM and (c) 900 RPM. All panels share the same axis scale.



(a) 300 RPM



(b) 600 RPM



(c) 900 RPM

Figure 5.6: Normalized pressure at all eleven sensors under vertical-axis shear turbulence, over the first 10 s of the developed-flow window, at grid speeds of (a) 300 RPM, (b) 600 RPM and (c) 900 RPM. All panels share the same axis scale.

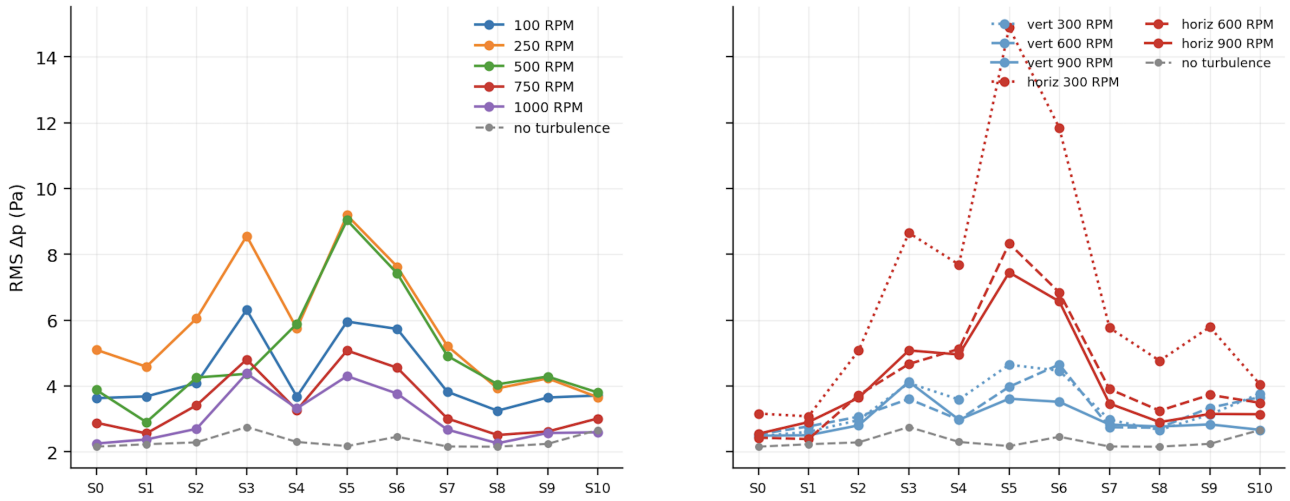


Figure 5.7: The RMS averages of each pressure sensor, on the left the pulsing turbulence modes is noted and on the right the two shear turbulence modes are seen.

6

Discussion

The results confirm the core of this project: a distributed array of low-cost barometric sensors can read the aerodynamic state of a bird-inspired airfoil. The clearest evidence is the chordwise pressure profile in Fig. 5.2, where the datasets show clear evidence for the creation of lift. The pressure sensors were able to determine that the shape would create lift even though it was fixed by tripods.

The project also verified that the the leading edge sensors captured most of the oncoming gusts. Fig. 5.7 shows exactly this. The strongest changes in amplitude are seen with sensors closest to the leading edge. The same shape is seen across p0 to p10, with a suction region near the leading edge on the upper surface and a recovery toward the 30 percent chord position. This is exactly the signature predicted by the pressure-coefficient picture in Section 2, and the fact that an eleven-sensor array reproduces it so cleanly is itself a validation that the taps were placed where the aerodynamically meaningful information lives. Sampling the leading third of both surfaces, as the literature study suggested, was therefore the right path.

Moreover for future drone designs, based on these findings, it can be said that most of the turbulence information was recorded by the sensors on the leading edge. [20] noted this and presented evidence that even a 3 sensor array captures the full aerodynamic profile. This evidence can also be used to state that sensors should be placed towards the leading edge of the sensors where the wing picks up most of the turbulence intensity.

A more subtle result hides in the spin-up trace of Figure 5.2. Lift does not appear as a step the instant the tunnel switches on but it builds gradually toward its developed value. It is tempting to read this as a physical echo of the Küssner build-up, where, when numerically solved, the lift coefficient climbs from zero and asymptotes as the wing immerses itself in the disturbance. The tunnel's fan spins up over a far longer timescale than a true sharp-edged gust, so the curve illustrates the principle of a progressive aerodynamic response rather than measuring a quick gust directly. Still, it shows the array is fast enough to resolve the transient development of lift, which is encouraging for any future gust work.

The turbulence comparison in Figure 5.4 indicates that the setup can distinguish between turbulent states. The convergence of some types of turbulence indicates however that the isolation of a turbulent state can be more challenging. Because the root mean squared equation was applied to the absolute pressure signal, which is dominated by the approximately 100 kPa atmospheric offset, the calculated RMS primarily reflects the mean pressure distribution rather than the pressure fluctuations associated with turbulence. A more representative metric is the RMS of the mean-subtracted pressure signal, which better captures the increased fluctuation amplitudes observed in the higher-energy pulsing and shear cases.

At the flight speed of $6.5\,\mathrm{m\,s^{-1}}$, the available dynamic pressure is approximately 26 Pa, which is comparable to the BMP280 sensor accuracy of $\pm 12\,\mathrm{Pa}$ [17]. Consequently, the pressure fluctuations produced by weak gusts are masked by the sensor's measurement uncertainty, indicating that the primary limitation of the system is pressure sensitivity rather than sampling rate, despite the sensor being capable of capturing sufficient samples within the gust response times reported by Cheney et al. This supports the conclusions of Mohamed et al. that pressure sensing is a promising modality for gust detection, while demonstrating that absolute barometric sensors such as the BMP280 lack the sensitivity required to resolve low-amplitude, transient gusts.

6.1 Design Limitations

While allowing for a large amount of aerodynamic information to be collected, the setup faced drawbacks. The nature of the BMP280 sensors provided slow sampling frequencies. with each measurement taking around 5 ms the phase advanced aspect of this research could not be properly tested with this apparatus. The setup had to be mounted using two adjacent tripods which restricted the wings rotation. Force data had to be interpreted by analyzing pressure differences rather than noticing the movement itself.

7

Conclusion

This report asked to what extent an array of basic barometric pressure sensors, distributed around the first 30% of the leading edge of a bio-inspired airfoil, can resolve variations in turbulence intensity in its surroundings. An Eppler 423 section was chosen for its high camber and its performance at low Reynolds numbers, 3D printed, and fitted with eleven BMP280 sensors, four on the bottom and seven on top, read out over an SPI bus by a Teensy 4.1. The wing was mounted in a wind tunnel at 6.5 ms^{-1} and a fixed angle of attack of 6 degrees, and exposed to still air, steady flow, and pulsing and shear turbulence at a range of grid speeds.

The array measures the aerodynamics of the wing. It registers the onset of wind and reproduces the pressure difference between the top and bottom surfaces that created the signature of lift. A steady state difference of about 23 Pa was recorded in Fig. 5.2 and the strongest response at the sensors nearest the leading edge. This is what the pressure distribution of Chapter 2 predicts, and it confirms that the taps are in a useful place.

On the question itself the answer is a qualified yes. The array separates smooth flow from turbulent flow clearly, and it separates the two shear cases from each other, with the disturbances that change the effective angle of attack giving the larger signal. This is what the hypothesis of Chapter 4 predicted. What the array does not do well is grade the turbulence within one type 5.6 5.5. The setup is therefore useful for recognizing the presence and the character of turbulence, but not for measuring how much of it there is.

Two things set that limit. The first is sensitivity. At 6.5 ms^{-1} the whole dynamic pressure is only 26 Pa, so a weak gust sits close to the noise of a single sensor. The BMP280 is built for altimetry, where the pressures of interest are hundreds of pascals and slow, and this application asks it to work at the other end of its range. The second is the setup rather than the sensor. The wing was clamped and the incoming gust was never measured independently, so the phase-advanced question in the title of this report is left open. At 100 Hz, the sampling rate is the thing that would prevent it.

7.1 Future Work

Future research should focus on creating pressure sensing setups with much higher sampling rates. The higher sampling rates will provide the microcontroller with turbulence information faster and give the drone enough time to respond, giving the drone a true 'fly-by-feel' sensing ability. Research should also go into determining how fast the pressure sensors can respond to the wing moving, rather than, what this experiment did, quickly changing the angle of the oncoming flow by introducing gusts. This would put the sensor in the situation it would actually meet in flight, where the wing rotates into the

gust while the gust is still passing over it. A wing mounted on a bearing rather than clamped between tripods would allow this, as the pressure signal and the resulting rotation could then be recorded at the same time and the delay between them measured directly. That delay is the number a fly-by-feel controller has to work with, and it is the one this setup could not provide.

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Supplementary Material

.1 Appendix 1.1 - Additional derivations

.1.1 Roll rate derivation

Parameter	Proportionality	Reasoning
Span (b)	b	Governing length scale
Wing area (S)	b^2	$S = c\,b \sim b \cdot b$ (constant aspect ratio)
Moment of inertia (I_{xx})	b^5	$I_{xx} \sim m\,b^2 \sim b^3 \cdot b^2$, with $m \sim b^3$
Velocity (U)	$b^{0.5}$	$U \sim \sqrt{mg/S} \sim \sqrt{b^3/b^2}$
Dynamic pressure (q)	b	$q = \frac{1}{2}\rho U^2 \sim (b^{0.5})^2$

Table 1: Scaling of the parameters governing roll damping with wingspan b, assuming geometric scaling with b as the single governing length scale. The non-dimensional roll damping coefficient C_{l_p} is taken as scale-invariant.[14]

The roll mode time constant is the inverse of the roll damping derivative,

$$\tau_r = \frac{1}{|L_p|} = \frac{2\,I_{xx}\,U}{q\,S\,b^2\,C_{l_p}}. \tag{1}$$

Substituting the proportionalities from Table 1 and retaining only the powers of b,

$$\tau_r \sim \frac{I_{xx}\,U}{q\,S\,b^2} \sim \frac{b^5 \cdot b^{0.5}}{b \cdot b^2 \cdot b^2} = \frac{b^{5.5}}{b^5} = b^{0.5}, \tag{2}$$

which yields

$$\tau_r \sim \sqrt{b}. \tag{3}$$

.1.2 The Küssner problem and the Duhamel integral and it's computation.

$$w_g = \begin{cases} 0 & \text{if } \bar{x} > Vt/c, \\ w_0 & \text{if } \bar{x} \leq Vt/c, \end{cases} \tag{4}$$

In the equality above, w_g denotes the gust-induced vertical velocity experienced by the airfoil, w_0 is the constant gust velocity behind the gust front, V is the freestream velocity, t is time, and c is the airfoil chord length. The nondimensional coordinate $\bar{x}$ is measured along the chord from the leading edge. $\frac{VT}{c}$ gives the chordwise position of the gust front at time t. The resulting lift coefficient is written as,

$$C_l(t) = 2\pi \left(\frac{w_0}{V}\right) \psi(s), \tag{5}$$

where $\psi(s)$ is the Küssner function. It function builds from zero and asymptotes to unity as $s \to \infty$. Combined with the Duhamel superposition integral, it gives the lift response to an arbitrary vertical

upwash field,

$$C_l(t) = \frac{2\pi}{V}\left(w_g(0)\,\psi(s) + \int_0^s \frac{dw_g(\sigma)}{dt}\,\psi(s-\sigma)\,d\sigma\right). \tag{6}$$

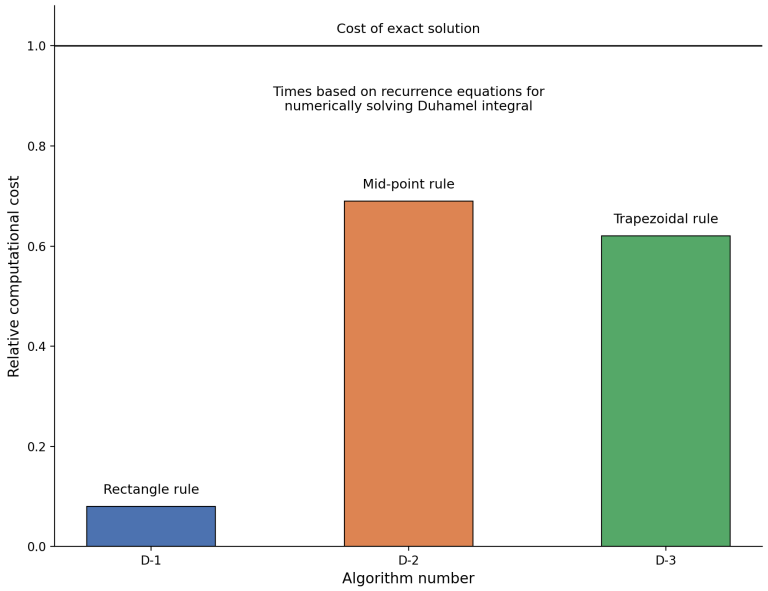


Figure 1: Relative computational costs of numerical solutions to the Duhamel integral using various numerical methods. Recreated from [13].

A convolution integral describes how a system's output is generated by combining an input signal with the system's response function at every instant in time. At each time step, the input history is multiplied point by point with a shifted, time-reversed response function. The resulting values are integrated to produce a single output. The response function is then shifted forward, and the process is repeated for the next time step. The relatively high computational cost comes from the repeated dependence on the full input history. Producing an accurate output at time t generally requires reprocessing the entire preceding signal rather than a single current value. N time steps the computation scales approximately as N^2. Numerical schemes such as the rectangle, midpoint, and trapezoidal rules approximate this process with varying degrees of accuracy.

The Küssner problem described in Section 2 requires using a Duhamel, or convolution integral, to be solved. This mathematical method cannot not be solved analytically and must be solved using a computational algorithm which naturally produces lag in terms of computational time. The computational cost of three suggested methods of solving such an integral are presented in Figure 1.

.2 Design Iterations

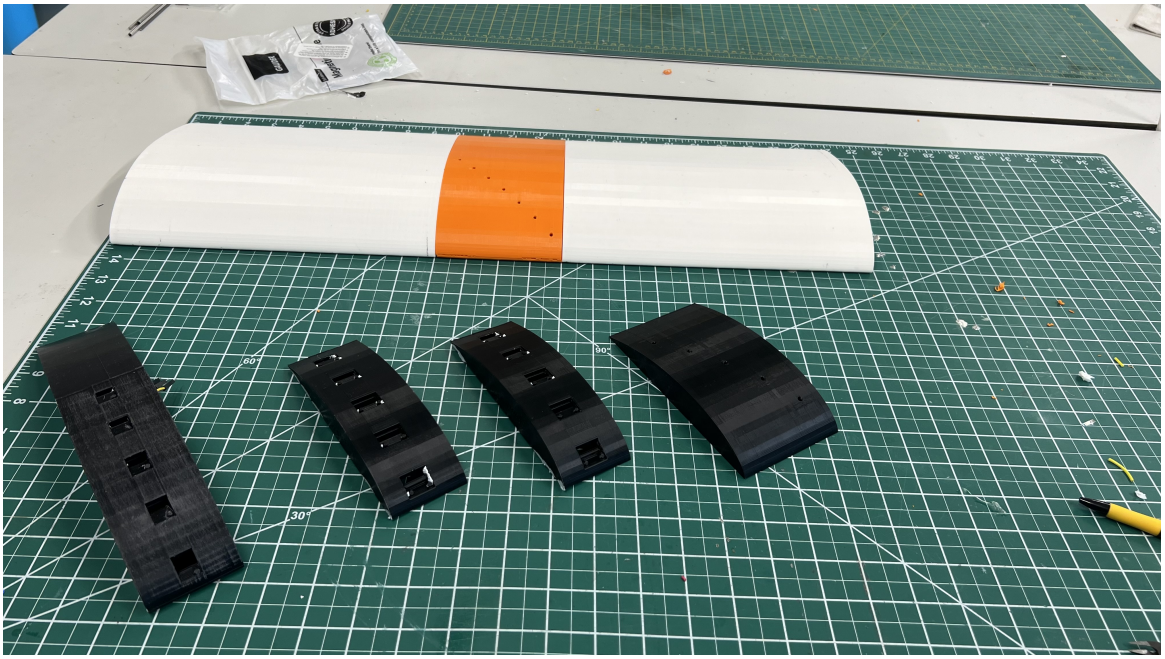


Figure 2: The various design iterations of the design. The black designs have open sensor placements and are aligned. This was found to create wake regions and not have a smooth aerodynamic shape and so the orange design was chosen.